

Automated Competitor Watch Report

Mini Project Guide — a hands-off weekly report on what your competitors are up to, built with sub-agents and delivered through Telegram every Monday morning

What you'll build

In this mini project you set up an automated competitor watch report. Your agent researches three of your competitors once a week, compiles a clean summary of what each one is doing, and delivers the report to your Telegram every Monday at 8am. You don't lift a finger. You just read the report when you wake up on Monday.

The project brings together several things you've learned so far in the course. Sub-agents handle the research in parallel so it actually finishes in a reasonable amount of time. The heartbeat mechanism handles the weekly schedule because Docker containers don't have system crontab. And the whole thing runs unattended once set up.

Use your real competitors. This guide shows the general pattern with example competitors like Zapier, Make.com, and n8n. When you set this up for yourself, swap those for your actual competitors in your industry. The value of this project comes from watching businesses that are genuinely competing with yours, not from watching random tools.

What you need before starting

Before you run this project, make sure you've covered these earlier lessons: Browsing the Web and Sub-Agents (so your agent can research online and spawn parallel workers), Cron Jobs and Heartbeat (so you understand how scheduled tasks work in OpenClaw), and Build Your Business Brain (so the agent has context about what your business does and what makes a piece of competitor info actually relevant to you).

You also need Brave Search or another web search provider configured in OpenClaw. Without web search, your sub-agents have nothing to research. Most of you will have this set up already from the Browsing the Web lesson.

The prompts used in this lesson

Here are all the prompts I used during the recording. Copy them as a starting point and adjust the competitor names and the focus areas to match your own business.

1. The main setup prompt

This is the prompt that kicks off the whole thing. It tells the agent who to research, what to look for, how to use sub-agents, and how to deliver the report.

PROMPT — Main setup

I want you to set up an automated weekly competitor watch report. Here is what I need. Every Monday at 8am, research these three competitors: Zapier, Make.com, and n8n. For each one, I want you to find out what they've launched recently, any major news or announcements from the past week, any pricing changes, and any notable social media activity or marketing campaigns. Use sub-agents to research all three in parallel so we don't wait forever. Then compile everything into one clean report with a section for each competitor. Send the final report to me here on Telegram. Run the first report now as a test so I can see what it looks like, and then set up the recurring Monday schedule after I confirm it looks good.

2. Fixing the crontab issue

During the recording, the agent first tried to use system-level crontab, which isn't available in Docker containers. When that failed, it suggested using OpenClaw's own heartbeat-based scheduling. This prompt tells the agent to go with the heartbeat approach and run a test right away.

PROMPT — Switch to heartbeat

Perfect, let's use OpenClaw's own heartbeat-based scheduler instead since we don't have system crontab. Set up the weekly report to run every Monday at 8am using the heartbeat mechanism. And go ahead and run a test report right now so I can see what it looks like before we enable the Monday schedule.

3. Adjusting the format (optional)

After the first test report comes in, you'll probably want to tweak the format. Maybe you want an executive summary at the top, or a specific ranking of which competitor moves matter most. Here's the kind of prompt that works for format changes.

PROMPT — Format change example

The report looks good but I want a few changes. Add a two-sentence executive summary at the very top that highlights the single most important move any competitor made this week. Then underneath that, keep the individual competitor sections as they are. Also add a section at the bottom called What This Means For Us that gives me a short take on how I should respond or adjust my own plans based on what the competitors are doing. Update the report format with these changes and use the updated version from next Monday onwards.

How the pieces fit together

It helps to understand what's actually happening behind the scenes when this report runs. Here's the full flow from Monday morning to Telegram delivery.

Step 1 — The heartbeat wakes the agent up

Every few minutes the agent checks whether there's anything scheduled to run. On Monday morning after 8am, it sees there's a competitor watch report due and starts the workflow. One thing to know here is that heartbeat-based scheduling isn't exact to the second. The report might go out at 8:02 or 8:05 depending on when the next heartbeat fires. For a weekly competitor report this doesn't matter at all, but it's worth knowing.

Step 2 — Three sub-agents start up in parallel

The main agent doesn't try to research all three competitors by itself, because that would take ages and risk hitting timeouts. Instead it starts up three sub-agents, one for each competitor. Each sub-agent is essentially its own independent session with its own context and its own research task. They all work in parallel. You can actually see them running if you open the gateway dashboard while this is happening.

Step 3 — Each sub-agent does its research

Each sub-agent uses web search to find out what its assigned competitor has been up to. It looks at the company's website, recent news, social media, and anything else relevant. Then it summarises its findings and hands the summary back to the main agent.

Step 4 — The main agent compiles the full report

Once all three sub-agents have reported back, the main agent puts everything into one clean document. One section per competitor, with a consistent structure so it's easy to scan. If you've added an executive summary or a What This Means For Us section, those get added here too.

Step 5 — Telegram delivery

The finished report lands in your Telegram chat with your agent. You open it Monday morning, scan through it, and you're caught up on your whole competitive landscape in about two minutes. No research meetings, no manual checks, no wondering whether you missed something important.

Why sub-agents matter for this project

If you tried to do this with a single agent running through each competitor one at a time, you'd have two problems. The first is speed — researching three competitors sequentially takes three times longer than researching them in parallel. The second is timeouts. A long research task running directly on the heartbeat can time out, and then the report just silently fails to arrive. With sub-agents, each one has its own context window and its own time budget. The main agent just coordinates and compiles. Nothing times out.

This is a pattern you'll come back to again and again once you get comfortable with it. Any time you have a scheduled task that involves multiple independent pieces of work, sub-agents are the right tool.

Troubleshooting

"Host does not have crontab"

If the agent tries to use system crontab and fails, that's normal for Docker-based setups. Just tell the agent to use OpenClaw's heartbeat mechanism instead. The example prompt for this is in the prompts section above.

Sub-agents take too long or hang

If one of the sub-agents gets stuck, the whole report can stall. Ask your agent to kill any hung sub-agent sessions and rerun the report. You can see active sessions in the gateway dashboard — the same one we covered in the Browsing the Web lesson.

Report arrives but the content is thin

If your first report is shorter than you'd like, the usual fix is to give the agent clearer guidance about what you want to see. Tell it what specific areas matter to you — pricing changes, feature launches, hiring patterns, content strategy — and it will dig deeper into those areas on the next run. The business brain you set up earlier helps a lot here because the agent already knows your industry and your focus.

Heartbeat fires too slowly for your liking

If the default heartbeat interval means your report arrives at 8:10 instead of 8:00, and you really do want it closer to the hour, you have two options. You can shorten your heartbeat interval so checks happen more often, or you can switch to Gateway cron which fires at exact times. For a weekly competitor report, the default heartbeat is fine. For something more time-sensitive, Gateway cron is worth the extra setup.

Ways to customise this project

Once the basic version is running and delivering to Telegram every Monday, you'll start thinking about what else you could do with the same pattern. Here are a few directions you can take it.

Watch more than three competitors

There's nothing special about the number three. If you have five or seven real competitors, use five or seven sub-agents. Just tell the agent to spread the research across however many parallel workers you need.

Change the focus for each competitor

You don't have to research every competitor the same way. Maybe for one competitor you care about pricing, for another you care about hiring, and for a third you care about content marketing. Tell your agent the specific focus for each one and the sub-agents will use that guidance.

Add a monthly deep-dive on top of the weekly summary

Keep the weekly report lightweight, and add a separate monthly deep-dive report that goes further — looking at longer trends, pricing history, product release cadence, and strategic shifts. This uses the same sub-agent pattern but with a different schedule and a different brief.

Link it to your Mission Control dashboard

If you've built the Mission Control dashboard earlier in the course, you can tell your agent to log each competitor report into a dedicated screen there. That way you build up a history over time and can scroll back to see what each competitor has been doing month over month.

Feed the findings into your own content

Once your agent knows what competitors are publishing, you can ask it to draft response content from your angle — a LinkedIn post, a blog outline, a sales email that positions you against a specific competitor move. This turns the competitor watch from an information-gathering exercise into an actual marketing asset.

Final note

Swap the examples for your real competitors. The Zapier, Make.com, and n8n examples in this guide are just placeholders so you can see the pattern. Once you understand how the pieces fit together, spend a few minutes thinking about who you genuinely compete with and set up your version accordingly. That's where the real value of this project kicks in. You're not trying to watch the automation industry — you're trying to know what the businesses closest to yours are doing, week after week, without having to remember to check.